A Multiscale 0D–1D Model of Valveless Pumping for Biomedical Applications

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Introduction

Valveless pumping based on the Liebau effect has attracted increasing interest as a practical mechanism to generate pulsatile flow without conventional valves, with relevance to extracorporeal circulation and other biomedical applications [1]. Several experimental studies have characterized the influence of actuator position, excitation frequency, tube compliance, and hydraulic asymmetry on the resulting flow [2,3].
However, the numerical simulation of such systems remains limited, particularly for configurations in which a local displacement-controlled pincher periodically deforms a compliant tube. This work develops a 0-D/1-D framework in which the displacement of the pincher is applied as a prescribed deformation over a finite segment of the compliant tube, while the vessel and reservoir dynamics are solved by the coupled model.

Methods

Figure 1 illustrates the schematic of the pumping configuration based on the Liebau effect. According to the figure, two reservoirs are connected through a system of compliant tubes with different properties.

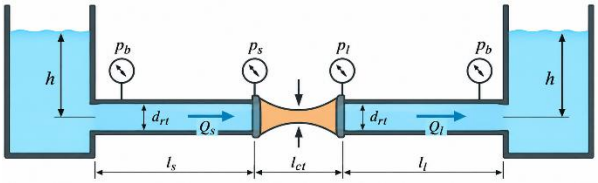


Figure 1: Schematic representation of the Liebau-pump system

The compliant tubes are described by the one-dimensional conservation equations as eqs. 1 and 2.

$$\frac{\partial A}{\partial t} + \frac{\partial Q}{\partial x} = 0 \qquad (1)$$

$$\frac{\partial Q}{\partial t} + \alpha \frac{\partial}{\partial z}\left(\frac{Q^2}{A}\right) + \frac{A}{\rho}\frac{\partial P}{\partial z} + K_r \frac{Q}{A} = 0 \qquad (2)$$

Closed by the pressure-area relation as eq. 3.

$$P - P_{ext} = E h_0 \sqrt{\pi} \frac{\sqrt{A} - \sqrt{A_0}}{A_0} \qquad (3)$$

The left and right finite reservoirs are modelled via $C_L \frac{dP_L}{dt} = -Q_L$ and $C_R \frac{dP_R}{dt} = +Q_R$ where $C_r = \frac{A_r}{\rho g}$ which are coupled to the 1-D model through the corresponding boundary pressures and flows.
In addition, each pumping cycle is divided into a compression phase and a free-recovery phase to mimic the actual behavior of the pincher. The pincher acts locally over a segment of the middle tube defined as: $\Omega_{act} = \left[x_c - \frac{L_{act}}{2}, x_c + \frac{L_{act}}{2}\right]$.

Results

Figure 2 shows the hydrostatic response of the symmetric configuration. Transient oscillations in reservoir height and head difference decay over time without producing sustained directional transfer, leaving the net transferred volume negligible and the reservoir levels effectively unchanged.

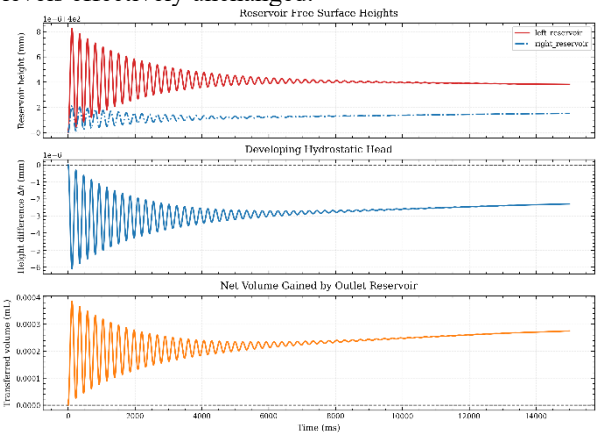


Figure 2: Hydrostatic response of a symmetric pumping configuration

Conclusions

A multiscale 0D-1D framework was developed to represent valveless pumping generated by deformation of a compliant tube. The model combines the 1-D flow model with finite-reservoir boundary conditions and a localized displacement-driven actuation strategy.
Future work will focus on a detailed parametric study to quantify the impacts of pincher location and width, compression characteristics, tube compliance, fluid properties, and hydraulic asymmetry on the flow and pressure response.

Acknowledgements

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References

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